Azure Cosmos DB SQL API Assignment — Hotel Booking Analytics

Create

```
Database: hotel_network
Container: bookings
Partition Key: /city
```

Data

Insert these documents into the `bookings` container.

```
{
  "id": "HB001",
  "guest_name": "Arjun Sethi",
  "city": "Hyderabad",
  "hotel_name": "Grand Meridian",
  "room_type": "Deluxe",
  "nights": 3,
  "room_rate": 5200,
  "booking_amount": 15600,
  "booking_status": "Confirmed",
  "rating": 4.5
}
```

```
{
  "id": "HB002",
  "guest_name": "Mira Kulkarni",
  "city": "Hyderabad",
  "hotel_name": "Lakeview Residency",
  "room_type": "Standard",
  "nights": 2,
  "room_rate": 3200,
  "booking_amount": 6400,
```

```
  "booking_status": "Completed",
  "rating": 4.1
}

{
  "id": "HB003",
  "guest_name": "Faizan Mir",
  "city": "Bengaluru",
  "hotel_name": "Tech Park Suites",
  "room_type": "Executive",
  "nights": 4,
  "room_rate": 6800,
  "booking_amount": 27200,
  "booking_status": "Confirmed",
  "rating": 4.7
}

{
  "id": "HB004",
  "guest_name": "Nisha Balan",
  "city": "Bengaluru",
  "hotel_name": "Urban Nest",
  "room_type": "Standard",
  "nights": 1,
  "room_rate": 3500,
  "booking_amount": 3500,
  "booking_status": "Cancelled",
  "rating": 3.9
}

{
  "id": "HB005",
  "guest_name": "Ritesh Mehra",
  "city": "Chennai",
  "hotel_name": "Marina Crown",
  "room_type": "Suite",
  "nights": 3,
  "room_rate": 9000,
  "booking_amount": 27000,
  "booking_status": "Completed",
  "rating": 4.8
}
```

```
{
  "id": "HB006",
  "guest_name": "Tara Nair",
  "city": "Chennai",
  "hotel_name": "Bay Comfort",
  "room_type": "Deluxe",
  "nights": 2,
  "room_rate": 4800,
  "booking_amount": 9600,
  "booking_status": "Confirmed",
  "rating": 4.2
}

{
  "id": "HB007",
  "guest_name": "Kabir Oberoi",
  "city": "Pune",
  "hotel_name": "Central Park Inn",
  "room_type": "Executive",
  "nights": 5,
  "room_rate": 6500,
  "booking_amount": 32500,
  "booking_status": "Completed",
  "rating": 4.6
}

{
  "id": "HB008",
  "guest_name": "Sneha Joshi",
  "city": "Pune",
  "hotel_name": "Green Leaf Hotel",
  "room_type": "Standard",
  "nights": 3,
  "room_rate": 3000,
  "booking_amount": 9000,
  "booking_status": "Confirmed",
  "rating": 4.0
}

{
  "id": "HB009",
  "guest_name": "Yusuf Merchant",
```

```
  "city": "Mumbai",
  "hotel_name": "Sea Crest",
  "room_type": "Suite",
  "nights": 4,
  "room_rate": 10500,
  "booking_amount": 42000,
  "booking_status": "Confirmed",
  "rating": 4.9
}

{
  "id": "HB010",
  "guest_name": "Divya Menon",
  "city": "Mumbai",
  "hotel_name": "Metro Grand",
  "room_type": "Deluxe",
  "nights": 2,
  "room_rate": 5500,
  "booking_amount": 11000,
  "booking_status": "Completed",
  "rating": 4.3
}

{
  "id": "HB011",
  "guest_name": "Aman Verma",
  "city": "Delhi",
  "hotel_name": "Capital Palace",
  "room_type": "Executive",
  "nights": 3,
  "room_rate": 7200,
  "booking_amount": 21600,
  "booking_status": "Confirmed",
  "rating": 4.4
}

{
  "id": "HB012",
  "guest_name": "Leena Francis",
  "city": "Delhi",
  "hotel_name": "North Gate Hotel",
  "room_type": "Standard",
```

```
  "nights": 2,
  "room_rate": 3400,
  "booking_amount": 6800,
  "booking_status": "Cancelled",
  "rating": 3.8
}

{
  "id": "HB013",
  "guest_name": "Vikram Rao",
  "city": "Jaipur",
  "hotel_name": "Royal Courtyard",
  "room_type": "Suite",
  "nights": 3,
  "room_rate": 8800,
  "booking_amount": 26400,
  "booking_status": "Completed",
  "rating": 4.7
}

{
  "id": "HB014",
  "guest_name": "Pooja Arora",
  "city": "Jaipur",
  "hotel_name": "Pink City Stay",
  "room_type": "Deluxe",
  "nights": 2,
  "room_rate": 4500,
  "booking_amount": 9000,
  "booking_status": "Confirmed",
  "rating": 4.1
}

{
  "id": "HB015",
  "guest_name": "Noel George",
  "city": "Kochi",
  "hotel_name": "Harbour View",
  "room_type": "Executive",
  "nights": 4,
  "room_rate": 6200,
  "booking_amount": 24800,
```

```
  "booking_status": "Completed",
  "rating": 4.6
}

{
  "id": "HB016",
  "guest_name": "Ishita Sen",
  "city": "Kochi",
  "hotel_name": "Palm Grove",
  "room_type": "Standard",
  "nights": 1,
  "room_rate": 2800,
  "booking_amount": 2800,
  "booking_status": "Confirmed",
  "rating": 4.0
}

{
  "id": "HB017",
  "guest_name": "Rahim Khan",
  "city": "Bhopal",
  "hotel_name": "Lake Square",
  "room_type": "Deluxe",
  "nights": 3,
  "room_rate": 4700,
  "booking_amount": 14100,
  "booking_status": "Confirmed",
  "rating": 4.2
}

{
  "id": "HB018",
  "guest_name": "Keerthi Iyer",
  "city": "Bhopal",
  "hotel_name": "Central Stay",
  "room_type": "Standard",
  "nights": 2,
  "room_rate": 3100,
  "booking_amount": 6200,
  "booking_status": "Completed",
  "rating": 4.1
}
```

Section A — Basic SELECT

1. Display all documents in the container.
2. Display only `guest_name`, `city`, and `hotel_name`.
3. Display `guest_name`, `room_type`, `nights`, and `booking_amount`.
4. Display `id`, `guest_name`, and `booking_status`.

Section B — WHERE Filters

5. Display all bookings from Hyderabad.
6. Display all bookings from Mumbai.
7. Display all bookings with status `Completed`.
8. Display all `Suite` bookings.
9. Display all bookings with room type `Executive`.

Section C — Numeric Filters

10. Display bookings where `booking_amount` is greater than `20000`.
11. Display bookings where `room_rate` is greater than `6000`.
12. Display bookings where `nights` is greater than `3`.
13. Display bookings where `rating` is below `4.2`.
14. Display bookings where `booking_amount` is between `8000` and `25000`.
15. Display bookings where `room_rate` is below `4000`.

Section D — AND Conditions

16. Display bookings that are `Confirmed` and have a booking amount greater than `15000`.

17. Display Mumbai bookings where the room type is `Suite`.
18. Display bookings where `nights > 2` and `rating > 4.5`.
19. Display Completed bookings where `booking_amount > 10000`.
20. Display Standard rooms where `room_rate < 3500`.

Section E — OR Conditions

21. Display bookings from Hyderabad or Bengaluru.
22. Display bookings from Mumbai or Delhi.
23. Display bookings where room type is `Suite` or `Executive`.
24. Display bookings whose status is `Completed` or `Confirmed`.
25. Display bookings from Chennai, Pune, or Kochi.

Section F — ORDER BY

26. Display all bookings ordered by `booking_amount` from lowest to highest.
27. Display all bookings ordered by `booking_amount` from highest to lowest.
28. Display bookings ordered by `rating` from highest to lowest.
29. Display bookings ordered by `nights` from highest to lowest.
30. Display bookings ordered first by `city` and then by `booking_amount`.

Section G — COUNT

31. Count the total number of bookings.
32. Count the number of Confirmed bookings.
33. Count the number of Completed bookings.
34. Count the number of Suite bookings.

35. Count the number of bookings in Pune.

Section H — GROUP BY

36. Count bookings by city.
37. Count bookings by `room_type`.
38. Count bookings by `booking_status`.
39. Find total booking amount for each city.
40. Find average booking amount for each room type.
41. Find average rating for each city.
42. Find total nights booked for each city.
43. Find average room rate for each room type.

Section I — GROUP BY with Filters

44. Find total booking amount by city using only Confirmed bookings.
45. Count Completed bookings in each city.
46. Find average booking amount by city for bookings above `8000`.
47. Find total nights grouped by room type.

Section J — Combined Queries

48. Find the top five bookings by `booking_amount`.
49. Find the top three bookings by `rating`.
50. Display Confirmed Suite bookings ordered by `booking_amount` descending.
51. Display bookings where `booking_amount > 15000` and `rating > 4.5`, ordered by booking amount descending.

52. Display Hyderabad or Bengaluru bookings where `nights > 2`.
53. Display Cancelled bookings ordered by rating from lowest to highest.
54. Display `guest_name`, `city`, `hotel_name`, `booking_amount`, and `rating` for bookings above `20000`.
55. Display Executive room bookings where `room_rate >= 6000`, ordered by room rate descending.

Section K — Business Questions

56. Write a query to determine which city has the highest total booking amount.
57. Write a query to determine which booking has the highest booking amount.
58. Write a query to determine which booking has the highest rating.
59. Write a query to determine which room type has the highest average booking amount.
60. Write a query to determine which city has the highest total number of nights booked.
61. Write a query to determine how many bookings are currently Cancelled.
62. Write a query to determine which city has the highest number of Completed bookings.
63. Write a query to determine which hotel has the highest booking amount.
64. Write a query to determine which room type has the highest average room rate.
65. Write a query to determine the total booking amount for all Confirmed bookings.